THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 08

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 5 units	$\sqrt{(4-(1))^2+(6-(2))^2}=\sqrt{9+16}=5$
2	**(a)** $5x-8$	degree 1
3	**(c)** $\mathbb{Q}$	Q for quotient
4	**(a)** $9x^2-24x+16$	$(a-b)^2=a^2-2ab+b^2$
5	**(b)** 10 cm	half-chord $=\sqrt{13^2-12^2}=5$, so chord $=2\times5=10$
6	**(a)** the centre of the circle	centre is equidistant from the chord's endpoints
7	**(c)** equal area	equal bases and the same height
8	**(d)** 12	multiply the number of outcomes of each experiment
9	**(b)** 38	substitute $n=7$
10	**(b)** 10 factors; it is not a perfect square	count the factor pairs; an odd count signals a repeated pair
11	**(a)** true	15 = 3 × 5 with 3 and 5 having no common factor
12	**(a)** 19%	$\dfrac{80\times25+120\times15}{80+120}=19\%$
13	**(d)** A is false but R is true	A is false (totals unknown); R is true
14	**(a)** 2	each reduction replaces (m, n) by (n, m mod n); ends at gcd(75, 0)
15	**(d)** 3	carries by column: [1, 1, 1, 0]
16	**(c)** 7 cm	PQ = BC/2
17	**(c)** 132°	adjacent angles of a parallelogram add up to 180°
18	**(b)** c = 0	x = 0, y = 0 satisfies ax + by = 0
19	**(b)** 1848 cm^3	$\pi r^2h=1848$
20	**(b)** 150 cm^2	side $=5$ cm, TSA $=6a^2=150$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	LHS $=27$; RHS $=8+12+6+1=27$ ✓.	1+1
22	The four points are concyclic (Theorem 10): they lie on one circle.	2
23	A median divides a triangle into two triangles of equal area (equal bases, same height): $\text{ar}(ABD)=24$ cm^2.	1+1
24	$a=3, d=5$: $t_{10}=3+9\times5=48$; $t_{26}=3+25\times5=128$.	1+1
25	25% of $800=\frac{25}{100}\times800=200$ units (2).	1+1
26	$17 \bmod 5=2$ and $33 \bmod 7=5$ (1). They are the remainders on division (1).	1+1
27	The diagonals bisect each other (1): $OA=7$ cm, $OB=5$ cm (1).	1+1
28	$\frac{3}{6}=\frac{2}{4}=\frac{1}{2}$ but $\frac{5}{11}\neq\frac{1}{2}$ (1). The lines are parallel, so there is no solution (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	$y=-2x$ (each y is -2 times x). Slope -2 ⇒ the line falls; y decreases by 2 for each unit increase in x (linear decay).	1+1+1
30	Profit $=2400+500=2900$ (1½). $\frac{2900}{9000}\times100\approx32.2\%$ (1½).	1½+1½
31	Number theory: If n is a multiple of 8, then it is a multiple of 4; converse false (12) (1½). Geometry: If a quadrilateral is a square, then all angles are equal; converse false (a rectangle) (1½).	1½+1½
32	$2\cdot3^{n-1}=486\Rightarrow3^{n-1}=243=3^5\Rightarrow n=6$.	2+1
33	Distances: $\sqrt{16+16}=4\sqrt{2}$ from each. The three points form an isosceles triangle with the apex at $(0,-1)$; base $=8$ and $64=32+32$, so it is also right-angled at the apex.	1+1+1
34	(i) $\frac{1}{3}\times12=4$ (1). (ii) $\frac{6}{14}=\frac{3}{7}$ (2).	1+2

35	For $a = b = c = 1$: $1 + 1 + 1 = 3$ while RHS $= 6$. So it is NOT an identity.	2+1
	OR: $(3a)^3 - (2b)^3 = (3a - 2b)(9a^2 + 6ab + 4b^2)$.	1+2
36	In a parallelogram opposite angles are equal: $\angle A = \angle C$ (1). In a cyclic quadrilateral $\angle A + \angle C = 180^\circ$ (1). So each is 90° ⇒ rectangle (1).	1+1+1
	OR: Let $AB = CD$. In $\triangle AOB$ and $\triangle COD$: $OA = OC$, $OB = OD$, $AB = CD$ ⇒ congruent (SSS) (2). So $\angle AOB = \angle COD$ (1).	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $(11, 2) \to (2, 9) \to (9, 2) \to (2, 7) \to (7, 2) \to (2, 5) \to (5, 2) \to (2, 3) \to (3, 2) \to (2, 1) \to (1, 1) \to (1, 0)$: Step 3 is applied 7 times (2). (ii) $(11, 2) \to (2, 1) \to (1, 0)$ (1). (iii) 7 versus 2 (1). (iv) Subtraction depends on the value, the division algorithm does not grow with the value (1).	2+1+1+1
	OR: Compute divisors(n); if the list has exactly the two entries 1 and n, report prime; else not prime (2). divisors(21) $= [1, 3, 7, 21]$ — 4 entries ⇒ not prime (1). divisors(23) $= [1, 23]$ ⇒ prime (2).	2+1+2
38	Cone $= \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$ (1). Sphere $= \frac{4}{3}\pi r^3$ (1). Cylinder $= \pi r^2(2r) = 2\pi r^3$ (1). Dividing by $\frac{2}{3}\pi r^3$: $1 : 2 : 3$ (1). For $r = 3$: $18\pi : 36\pi : 54\pi = 1 : 2 : 3$ (1).	1+1+1+1+1
	OR: New radius $= 1.2r$ (1). (i) CSA $= 2\pi rh$ becomes $1.2\times$: increase 20% (1). (ii) Volume $= \pi r^2 h$ becomes $1.44\times$: increase 44% (1). (iii) With $h = r$: old TSA $= 2\pi r(2r) = 4\pi r^2$; new TSA $= 2\pi(1.2r)(r + 1.2r) = 5.28\pi r^2$, an increase of 32% (2).	1+1+1+2
39	(i) $(1100 + 4)^2 = 1210000 + 8800 + 16 = 1218816$. (ii) $(200 - 2)^2 = 40000 - 800 + 4 = 39204$. (iii) $(a + b)^2$ for (i), $(a - b)^2$ for (ii).	2+2+1
	OR: (i) $x^3 - y^3$ and $x^3 + y^3$. (ii) $(3p + 2q)(9p^2 - 6pq + 4q^2)$; $(4 - a)(16 + 4a + a^2)$.	3+2
40	(i) Points $(1, 1), (2, 5), (3, 9), (4, 13), (5, 17)$ lie on the line $y = 4x - 3$ (2). (ii) $(1, 3), (2, 6), (3, 12), (4, 24), (5, 48)$ do not lie on a line; slopes 3, 6, 12, 24 keep rising (2). (iii) An AP grows by a fixed amount (linear), a GP by a fixed multiple (much faster) (1).	2+2+1
	OR: (i) $\frac{n(n+1)}{2} + \frac{n(n-1)}{2} = \frac{n(2n)}{2} = n^2$ (2). (ii) $55 + 45 = 100$ ✓ (1). (iii) $t_{17} = 153$, $t_{80} = 3240$ (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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